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Development of a Hydrogen Micro Gas Turbine Combustor: Atmospheric Pressure Testing

Tom Tanneberger[1]
Chair of Fluid Dynamics,
Technische Universität Berlin,
Müller-Breslau-Str.8,
Berlin 10623, Germany
e-mail: Tom.Tanneberger@tu-berlin.de

Johannes Mundstock
Euro-K GmbH,
Burger Chaussee 25,
Berlin 03044, Germany

Christoph Rex
Euro-K GmbH,
Burger Chaussee 25,
Berlin 03044, Germany

Sebastian Rösch
Euro-K GmbH,
Burger Chaussee 25,
Berlin 03044, Germany

Christian Oliver Paschereit
Chair of Fluid Dynamics,
Technische Universität Berlin,
Müller-Breslau-Str.8,
Berlin 10623, Germany

The vision of a carbon-neutral world implies the shift from fossil to clean fuels for combustion-driven processes and machines like gas turbines. Green hydrogen is a promising alternative to substitute natural gas and other fossil fuels. In the H2mGT project, funded by the German BMWK, a microgas turbine (mGT) burner with 100% hydrogen firing is developed and validated. The project is collaboration between Technische Universität Berlin (TUB) and the manufacturer Euro-K GmbH. The project consists of three phases: (1). Atmospheric pressure tests with a fused silica combustion chamber; (2). Atmospheric pressure tests with counterflow-cooled steel combustion chamber and secondary air injection; (3). Validation of the burner in the micro gas turbine at elevated pressure levels. This paper will present the results of Phase 1. The hydrogen burner is based on a swirl-stabilized burner of TUB and was scaled to match the requirements of the mGT with its 130 kW thermal power. The burner design features multiple geometrical parameters to enable the optimization of the flame towards low NO_x emissions. Therefore, a variable swirl intensity, additional axial momentum of air in the mixing tube, a movable center-body and different fuel injection locations are implemented. Phase 1 investigates the parameter space in terms of flame stability, operational range, and parameter impact on flame shape and emissions. Therefore, temperature, pressure, and emission measurements as well as OH imaging are carried out. It is found that the flame can be operated over a large range of equivalence ratios and preheating temperatures up to 500 °C for many parameter settings. However, at some configurations, flashback into the mixing tube is triggered. As expected, the NO_x emissions are mainly influenced by the equivalence ratio, the fuel distribution, and the swirl intensity. Single-digit emissions are reached up to an equivalence ratio of 0.4 at atmospheric pressure conditions. Furthermore, at low air mass flow, the burner can be operated at 100% natural gas or 100% hydrogen without any geometry changes. The fuel switch, thereby, does not change the NO_x emissions significantly if reasonable normalization is used.* [DOI: 10.1115/1.4063708]

Keywords: micro gas turbine, combustion, hydrogen, emissions

1 Introduction

The current developments in the global climate crisis make a rapid shift away from fossil fuels inevitable. Nevertheless, combustion engines—especially gas turbines—will continue to be an essential part of our energy supply thanks to their flexibility and reliability, but fossil fuels need to be replaced. A CO_2-free alternative is green hydrogen produced from renewable energies. The joint H2mGT project of the TUB and the micro gas turbine manufacturer Euro-K GmbH aims to develop a hydrogen (H_2) combustor for a micro gas turbine.

The topic of H_2 combustion has now also arrived in the gas turbine industry. Almost all major manufacturers, such as Siemens [1–7], GE [8–10], Mitsubishi Heavy Industries [11–13], Ansaldo Energia [14–16], Solar Turbines [17] and PSM [18], are developing various combustion systems for syngas and H_2-enriched natural gas. In general, increased NO_x emissions and problems with flame stability are reported, so the H_2 content in the fuel remains limited. Very few gas turbines are suitable for 100% H_2 so far. One was commissioned in 2010 in Fusina, Italy (modified GE10.1 with 12 MW), another with 1 MW was tested by Kawasaki Heavy Industries in 2018 in Kobe, Japan [19]. A recent overview about the developments in the industry is given by the ETN Global's Hydrogen Gas Turbines report [20].

On the lab-scale level, the flame stability and the emissions of mixtures with 45%$_{vol}$ H_2 in natural gas have been investigated by Schefer et al. [21,22]. H_2 enrichment increases the risk of flashback but improves the lean blow-off limit, enhancing the flammability range. They found reductions in CO concentration with H_2 addition, without adversely affecting the NO_x emissions. Hussain et al. [23] measured a drastic increase in NO_x levels for H_2 addition of 30%$_{vol}$ in methane and ethylene flames. However, the H_2 was injected separately and further downstream than the main fuel, which leads to a much higher stratification. The flashback propensity was studied by Beerer et al. [24], Page et al. [25], and Syred et al. [26]. They also

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[1]Corresponding author.
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measured turbulent flame speeds at elevated pressures and temperatures. In general, numerous studies showed that fuels with high H_2 contents reveal higher flashback propensity as well as increased NO_x emissions [27–29].

A perfect platform to test H_2 combustion on a machine level is the mGT since the necessary investment, the operating pressure, and the exhaust gas temperatures are much lower than for heavy-duty gas turbines. In addition, mGTs usually offer good accessibility to the combustion chamber. They provide an alternative to gas piston engines in the small and medium combined heat and power segment. They are very flexible and emit significantly less pollutants than piston engines [30]. Modern mGT combustion chamber systems achieve NO_x emissions of 15 ppm, while current lean-burn engines in the medium power class with oxidation catalyst emit NO_x values of 125–250 ppm. The German Aerospace Center (DLR) [30] predicts a large potential for efficiency improvements and a rising share on the combined heat and power market for mGTs. In fact, there is a recent research interest in mGTs. However, most of the studies are done numerically. A review of such studies with respect to different mGT-based energy systems is given by Reale and Sannino [31]. On the experimental side, mGT combustors have been investigated for a variety of unconventional fuels thanks to their fuel flexibility. Liu et al. [32] studied the influence of different biofuels in an mGT combustor and found that the NO_x emissions increase with the fuel's Wobbe Index. Furthermore, the fuel injection method has a significant influence on the pollutant emissions. Okafor et al. [33] measured a strong rise of NO_x emissions for ammonia-enriched methane with single-stage combustion. In a small mGT (1–3 kW_{el}), du Toit et al. [34] measured emissions of $H_2/CH_4/CO_2$ fuel blends. Adding 15$\%_{vol}$ H_2 to CH_4 showed no significant effect on the NO_x emissions, while a 20/65/15$\%_{vol}$ mix reduced it by almost 50%. The study also reports the influence of the fuel switch on the mGT operation and auxiliaries. On a larger mGT (Capstone C30, 30 kW_{el}), de-Santoli et al. [35] tested H_2 addition in natural gas up to 10$\%_{vol.}$ In this study, the H_2 was used to compensate the engine derating due to increased inlet temperature in the summertime. A slight increase of the NO_x is reported for the H_2 enrichment. The authors also discuss the problem of the correct normalization of the emissions measured. As a result, they chose specific emissions instead of volumetric ones to evaluate the environmental impact of the mGT. In the current study, the normalization issue is discussed in the last section. The UCI Combustion Laboratory retrofitted a Capstone C60 mGT [36,37] for the operation with H_2. Despite adjustments to the fuel injection, the tests showed significantly increased NO_x emissions. The integrated combustion chamber of this machine allows only a few changes in this respect. In contrast, a silo with a single can combustor is used in the current project, which features better accessibility. Devriese et al. [38] also found high NO_x emissions for a mGT burner with H_2 firing in numerical studies.

Cappelletti et al. [39] numerically studied the burner of a T100 mGT (100 kW_{el}) and found that the original design is not suitable for 100% H_2, as the risk of flame flashbacks is too high. The issue of flashback was also addressed by Pappa et al. [40]. They showed by LES that flashback occurred already at 10$\%_{mass}$ H_2 enrichment in an unmodified T100 mGT combustor. As a countermeasure, humidified air (10$\%_{mass}$) was used to lower the reaction rate and, hence, prevent flashback. The concept of wet combustion was also previously studied at a similar burner as in the current study. The results show a strong reduction in NO_x emissions even at large H_2 contents and elevated pressure [41].

In summary, there are many challenges in the combustion of H_2. On the one hand, H_2 has a lower volumetric energy density (Table 3), which requires a higher fuel flow. On the other hand, H_2 has a high gravimetric energy density and a significantly higher reactivity compared to natural gas, which might increase the flame temperature and speed. The former can have a negative impact on NO_x emissions, while the latter makes it difficult to stabilize a premixed flame. In particular, the question of whether and to what extent H_2 increases NO_x emissions seems to be contradictory in the literature. These challenges are to be tackled in the present project with a swirl-stabilized dry low NO_x burner with axial air injection. The TUB has investigated a similar burner in numerous projects with H_2 and enriched natural gas and was able to achieve very good results regarding the prevention of flame flashbacks and the minimization of NO_x emissions [42–49]. The intended burner design is presented in detail in the first section.

The H2mGT project is divided into three phases. First, the base combustor is scaled and redesigned according to the requirements of the mGT. It is then tested in an atmospheric combustion test rig with a fused silica combustion chamber using OH* flame recording and emission analysis. In phase two, the transparent chamber is replaced by a steel flame tube with counterflow cooling and secondary air injection. Using this setup, the burner is optimized in the test rig regarding emissions and stable operation. After this phase is completed, the burner is adjusted for the integration in the mGT at Euro-K's test facility. Here, the pressure is increased and the new burner is validated under nominal operating conditions. The current paper presents the results of the project's first phase regarding stability range, flame shapes, and emissions. It additionally discusses the aspects of natural gas/hydrogen mixtures and flashback recovery.

2 Burner and Test Rig Design

The burner designed for this project is based on a generic swirl-stabilized burner, which was already extensively studied at TUB [42–49]. It was scaled to the thermal power of the micro gas turbine by maintaining similar volumetric flows in the burner. Geometrical parameters in the burner design are supposed to allow for an optimization of the operation range and NO_x emissions.

Figure 1 shows a schematic of the burner in Phase 1. The incoming air is preheated up to 500 °C and fed to the plenum via four inlets. In the plenum, the air flow splits between a tangential (via the swirler) and an axial flow to the mixing tube (left orange arrows in Fig. 1). As they are fed by the same plenum, the splitting ratio is solely determined by the pressure loss of each flow path. Both flows, the axial and the swirled, merge again in the mixing tube ($D_{mt} = 31$ mm) and enter the combustion chamber ($D_{cc} = 100$ mm, $L_{cc} = 180$ mm) further downstream. A third but much smaller flow of air is added to the mixing tube through a ring of holes to lean out the boundary layer.

The fuel can be injected through three different injection points (blue arrows in Fig. 1): (1). 16 nozzles ($D_{H_2} = 1.4$ mm) at the bottom of the mixing tube, which is regarded as technically premixed, (2). an axial nozzle ($D_p = 3$ mm) fuel lance ($D_{cb} = 6$ mm) in the mixing tube as central injector, (3). eight pilot nozzles ($D_{p_2} = 1$ mm) at the burner front plate around the mixing tube exit. The latter are supplied by a cavity behind the burner plate and can also be used with air for active or passive cooling of the plate. The front plate pilot is intended as emergency backup for flashback recovery. The idea of the central injection via the fuel lance is that the H_2 fraction profile in the mixing tube can be altered.

The burner features four parameters to alter the flow field and fuel distribution. These parameters are depicted in Fig. 1 and can be adjusted manually from outside without disassembling the burner. Hence, the design points in an optimization can be set during the operation of the burner, which allows for the evaluation of much more configurations. The parameters are

- S—swirler blockage in mm,
- A—axial injection restriction in # of blocked channels,
- L—length of the center body in mm, and
- P—share of fuel mass flow through fuel lance nozzle in %.

The swirler consists of eight tangential slots ($L_s = 27$ mm), which can be blocked partially (S). The axial air injection flows from a small preplenum through an annulus around the center body into the mixing tube. The preplenum is fed by eight radial channels from the main plenum. These channels can be separately blocked (A). The fuel lance is movable in the axial direction (L) and can be positioned between 0 mm and 90 mm, the latter corresponding to the end of the

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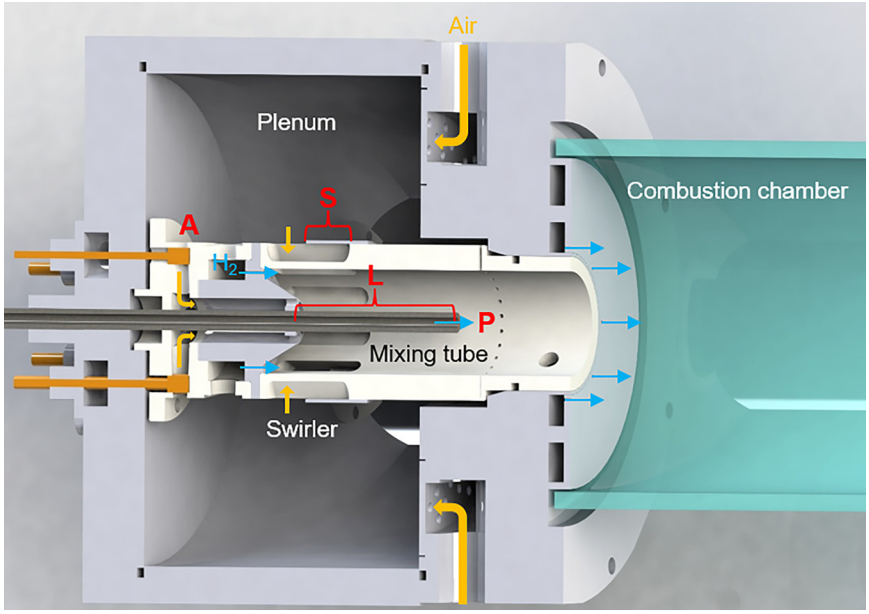

Fig. 1 Schematic of the burner with fused silica combustion chamber

mixing tube. The variation of this position has two effects. On the one hand, it changes the flow field in the mixing tube and the combustion chamber, because the lance body creates a wake in the flow. On the other hand, the point of the central fuel injection can be moved downstream leading to a stronger impact on the fuel distribution in the combustion chamber. Up to 50% of the fuel mass flow can be redistributed to the fuel lance (P). Both fuel injectors are supplied from a H_2 bundle via two individual mass flow controllers.

In Phase 1 of the project, the setup features a fused silica combustion chamber to gain optical access to the flame. The flame shapes are recorded in terms of the line-of-sight OH^* intensity by an intensified CCD camera with $310/\pm 10$ nm bandpass filter. The flame is ignited by a small natural gas ignition flame in the front plate, which will be replaced by a spark plug in the next phase of the project. Four thermocouples at the end of the combustion chamber measure the exhaust temperature at different radii, before the hot gas stream enters a water-cooled exhaust tube, releasing the gas into an open extraction unit. A small portion of the hot gas is sampled downstream of the combustion chamber by a one-hole probe. A gas analysis unit measures the NO_x content (± 1 ppm) for emission evaluation.

3 Measurement Settings

The micro gas turbine has a thermal power of 127 kW and a nominal air mass flow of 1280 kg/h. However, these values must be scaled down since the tests are conducted at atmospheric pressure. The thermal power was fixed to 36 kW for most of the experiments. In the mGT, the total air mass flow at baseload conditions is fixed by the output of the compressor, but the air is split into primary and secondary air. Only the primary air flows through the burner, while the secondary air is injected to the combustion chamber downstream of the flame. By changing the splitting ratio (e.g., changing the hole pattern in the combustion chamber wall), the local equivalence ratio of the flame can be varied. As there is no secondary air injection in the Phase 1 experiments, the total air flow $\dot{m}_{air}$ runs through the burner. Hence, $\dot{m}_{air}$ is varied to find an optimal primary equivalence ratio for the operation of the burner. The bounds are set to $\phi = 0.2$ and 0.6 to avoid blow off and high NO_x emissions, respectively. The air is preheated to 500 °C in the plenum to match the conditions of the recuperated mGT. The fuel is supplied at room temperature. However, the fuel heats up while flowing through the hot plenum. The actual fuel temperature of the premixed nozzles is measured by a thermocouple at the location of the H_2 label in Fig. 1 and is shown exemplarily in Fig. 2. The operating conditions are summed up in Table 1.

The mass flows and temperatures of a typical measurement series are shown in Figs. 2 and 3. Please note that at $\phi = 0.2$ and $\dot{m}_{air} = 180$ kg/h, the preheating temperature of 500 °C cannot be reached due to the limited power of the electric air heater. At high exhaust temperatures, the thermocouples start glowing. Hence, the exhaust temperatures had been corrected for the radiative heat loss using a model proposed by Príkopský [50]. The adiabatic flame temperature is calculated from the chemical equilibrium of the mixture using Gri-Mech 3.0.

Figure 4 shows that the exhaust gas temperature profiles at the end of the combustion chamber are very uniform, except close to the wall. The required turbine inlet temperature of the machine is around 840 °C, which is lower than the measured temperatures. However, the tests were conducted without secondary air that cools down the exhaust. Please note that the secondary air flow would increase with the equivalence ratio because the primary air is reduced (see Fig. 2).

4 Flow Field

The flow in the burner and the combustion chamber was studied by water tunnel particle image velocimetry and Reynolds-averaged Navier–Stokes (RANS) simulations (see Ref. [46] for methods). Some insights can be found in Tanneberger et al. [51] and a sample flow field is depicted by the axial velocity in Fig. 5.

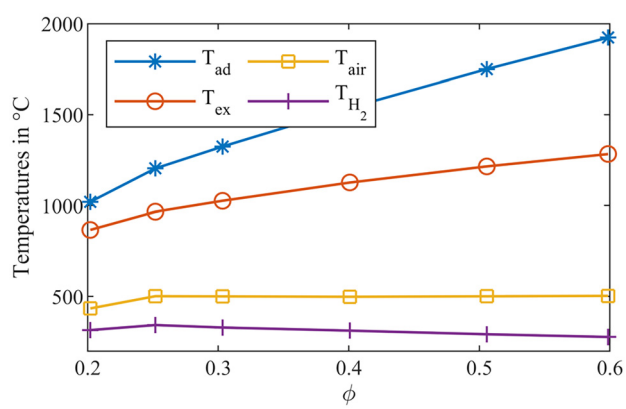

Fig. 2 Adiabatic flame, exhaust, air plenum and H_2 injection temperature of a measurement series

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Table 1 Operating conditions of the laboratory tests

Property	Symbol	Value	Unit
Thermal power	P_{th}	36	kW
H_2 mass flow	$\dot{m}_{H_2}$	1.08	kg/h
Air mass flow	$\dot{m}_{air}$	52.7–184.6	kg/h
Equivalence ratio	ϕ	0.2–0.6	
H_2 temperature	$T_{H_2,in}$	20	°C
Air temperature	T_{air}	500	°C

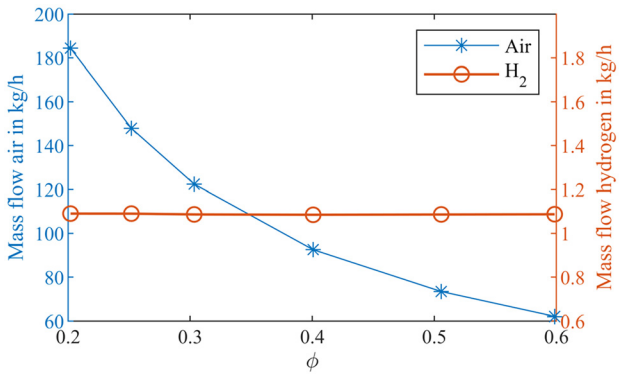

Fig. 3 Air and H_2 mass flows of a measurement series

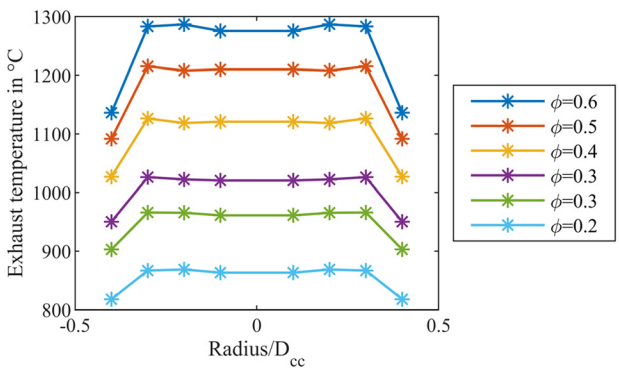

Fig. 4 Exhaust temperature profiles of a measurement series

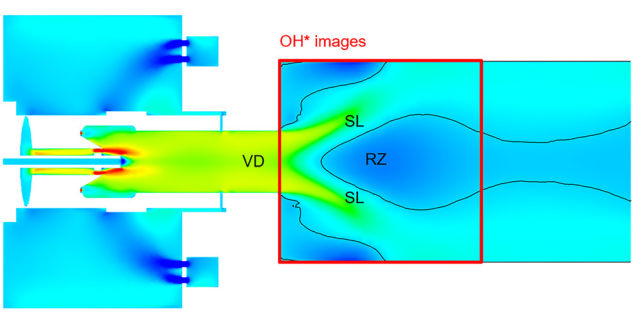

Fig. 5 RANS flow field (axial velocity and stagnation line) of the burner and combustion chamber. The frame on the combustion chamber shows the location of the following OH* images.

As the flow in the mixing tube is swirling, it induces a velocity deficit (VD) at the centerline of the mixing tube as seen in Fig. 6, which can increase the risk of flashback. At the area jump from the mixing tube to the combustion chamber, the swirled flow undergoes vortex breakdown and forms an inner recirculation zone (RZ) with negative axial velocity. The RZ is indicated by the black stagnation

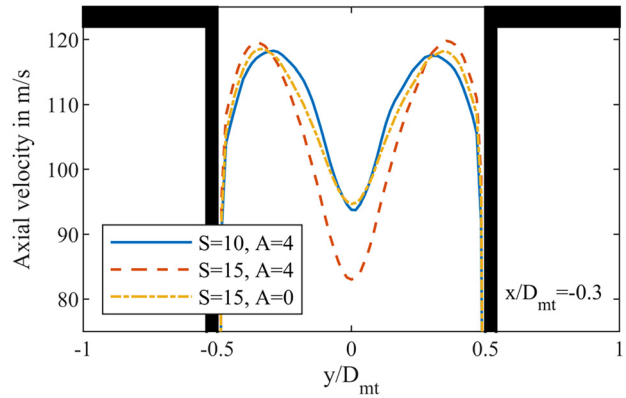

Fig. 6 Flow profiles in the mixing tube from RANS simulations

line in Figs. 5 and 7. Between the RZ and the outer recirculation zone, there is a high velocity jet transporting the fresh air/fuel mixture and generating strong shear layers (SL). These SL are intended to stabilize the flame.

The blockage of the swirler (S) increases the swirl intensity in the mixing tube and, therefore, leads to the upstream shift of the RZ shown in Fig. 7. Moreover, the VD is amplified by the enlarged swirl intensity. The latter effect can be partly compensated by increasing the axial air injection as shown by the flow profiles in Fig. 6. Opening the axial air injection channels (A) reduces the VD and leads to a small downstream shift of the RZ's stagnation point.

The fuel lance (L) has an additional effect on the VD due to its wake. If it is placed far downstream in the mixing tube, its wake can couple with the RZ such that the stagnation point shifts into the mixing tube. The wake effect can be compensated by activating the fuel flow through the central injector (P).

In general, the P injection allows for a variation of the fuel distribution at the exit of the mixing tube. Left side of Fig. 8 shows that the fuel is distributed homogeneously over the diameter of the mixing tube if no central fuel injection is applied ($P = 0\%$). At high injection rates ($P \geq 30\%$), the concentration profile shifts to a parabolic shape with a maximum in the center. The right side of Fig. 8 shows that the fuel concentration maximum shifts outside to the shear layers, if the fuel lance is introduced further into the mixing tube. This shift is driven by the flow field changes (RZ, VD) described before. At $P = 0\%$, this effect is rather small but it is amplified by increasing P. Please note that these concentrations have been measured in a water tunnel, not in the combustion test rig. Therefore, they give only qualitative insights into the mixing field and do not allow the calculation of local equivalence ratios. The water tunnel tests were carried out analogously to Reichel et al. [45]

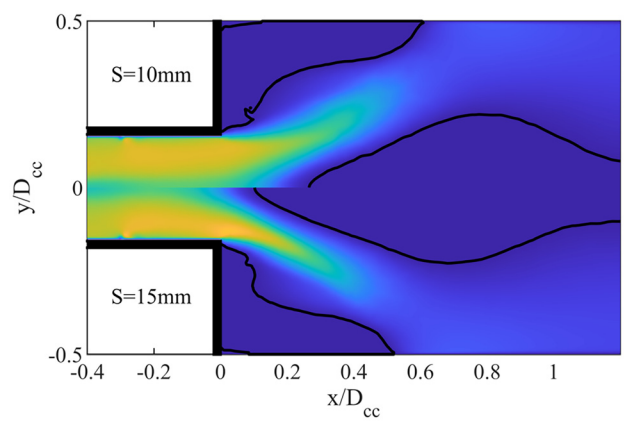

Fig. 7 RANS flow field (axial velocity and stagnation line) of the mixing tube and the combustion chamber for to different swirler blockage

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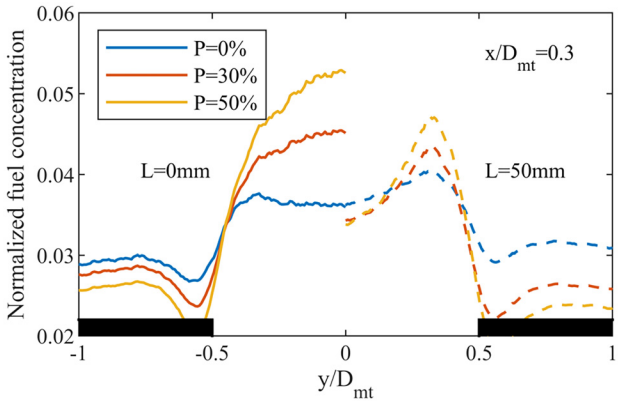

Fig. 8 Fuel distribution 10 mm downstream of the mixing tube exit measured in a water tunnel (solid $L = 0$ mm, dashed $L = 50$ mm)

with constant momentum ratios. Please refer to this publication for a description of the setup and a justification for the procedure.

5 Flame Shape and Stability Range

To conduct the optimization process in Phase 2 of the project, a map of stable operation points must be provided. Therefore, flame shapes have been recorded for a large number of parameter variations.

In the case of a stable flame, the technically premixed H_2/air mixture burns in V or M shape downstream of the mixing tube as shown in Fig. 9 (l.). However, geometry changes can alter the flow field in the aforementioned ways, which may trigger flashback as seen in the right image. Then, the flame propagates upstream into the mixing tube. The flame stabilizes itself in the wake of the center body and keeps burning as a jet flame in the mixing tube and the combustion chamber. In this case, the fuel valve is closed to avoid damage to the burner.

Figure 10 shows the operational range of the flame depending on three of the parameters. The different markers symbolize increasing levels of fuel share via the fuel lance (P). Hence, points with all five markers feature stability for a variety of fuel distributions and are very robust against flashback. The main driver of instability is the positioning of the fuel lance (L). For $L <= 10$ mm all flames are stable, while for higher L, the stability declines with increasing swirl intensity as shown by the dashed line in Fig. 10. Next to the flow field changes (VD and RZ), L varies the fuel distribution as seen in Fig. 8 and thus the local equivalence ratio and flame speeds. However, these changes in the concentration field are smaller than for a variation of P. Since the receptivity of flashback is still higher for L than for P, it can be concluded that the flashback is dominantly triggered by hydrodynamic effects.

Some points close to the stability line show an interesting behavior. They are marginally stable for $P = 0\%$ and $P \geq 40\%$. In

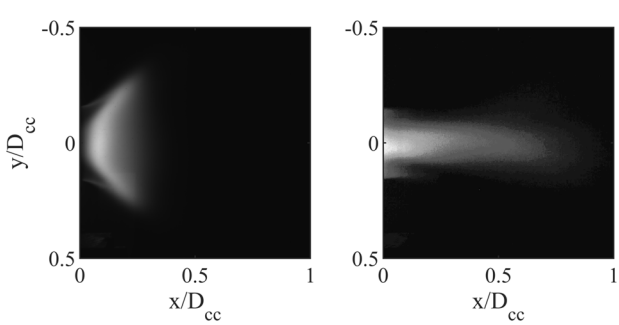

Fig. 9 OH* intensity (line-of-sight) of a stable flame (l.) (ϕ=0.4, $S = 10$ mm, $A = 0$, $L = 10$ mm, and $P = 0\%$) and flashback (r.)

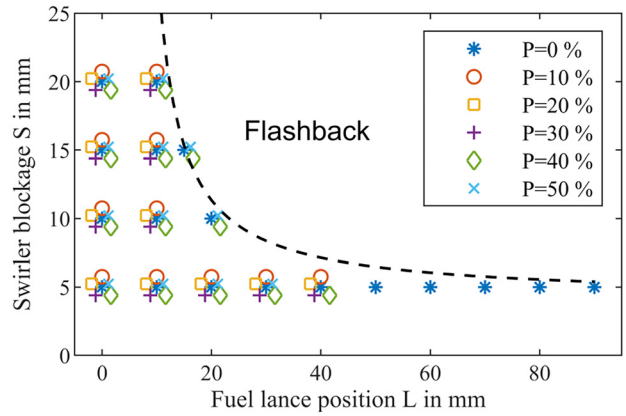

Fig. 10 Stable operating points for variations of the parameters S, L, and P at ϕ=0.4 and $A = 0$

between these stable bounds, the flame undergoes flashback. On the one side, shifting fuel to the central injection increases the local equivalence ratio near the centerline in the mixing tube. Thus, the flame speed increases and the flame propagates upstream along the VD. On the other side, if the fuel flow from the fuel lance is large enough, the VD is reduced by the axial fuel momentum and the flame is pushed back into the combustion chamber.

By applying a very low swirl intensity ($S = 5$ mm), it is possible to move the lance ($P = 0\%$) until the end of the mixing tube without triggering flashback. However, the flames in these configurations are not very robust and a variation of P has not been tested for $L \geq 50$ mm. Moreover, placing the fuel lance at the end increases its thermal load because the flame burns in the shear layers of the fuel lance (see Fig. 11(*d*) r.). Thus, configurations with $L > 70$ mm should be avoided.

The influence of the restriction of the axial air injection (A) is not shown in Fig. 10. Closing channels generally decrease the stability of the flame because the VD is amplified. However, the actual point of flashback depends strongly on the swirl intensity and fuel lance position. For example, six of the eight channels can be closed ($A = 6$) at $S = 10$ mm and $L = 10$ mm before flashback appears.

The utilization of the burner in the micro gas turbine does not only require static but also dynamic stability. During the experiments, the sound pressure levels had been monitored by a microphone in the exhaust tube. There are no indications of thermoacoustic instabilities under normal conditions at any of the tested parameter settings.

6 Parameter Study

This section presents an overview of the parameter's influence on the flame. The optimization in the subsequent phase of the project will be done with a constant splitting ratio between primary and secondary air, e.g., a nearly constant equivalence ratio. In order to set the equivalence ratio properly for the optimization, its influence on the flame is studied in the following paragraph.

The equivalence ratio is varied as described in Sec. 3 with a constant thermal power. It influences the flame in two ways, chemically and hydrodynamically. Increasing the equivalence ratio raises the adiabatic flame temperature (Fig. 2) and the laminar flame speed significantly. While the first increases the NO_x emissions (Fig. 12), the second might enhance the flame propagation against the main flow leading to an upstream shift of the flame or flashback in the worst case. The hydrodynamic effect, on the other hand, is based on the increased ratio of axial fuel momentum to circumferential air momentum at high equivalence ratios, which stabilizes the flow in the mixing tube. Reichel et al. [44] showed at a similar burner that this effect can lead to a higher robustness against flashback at high equivalence ratios than at low ones, which seems counterintuitive at first. The actual influence of the equivalence ratio on the flame in the current burner is shown in Fig. 11(*a*) by the deconvoluted (inverse Abel transformed) OH* intensity images. It can be seen that the

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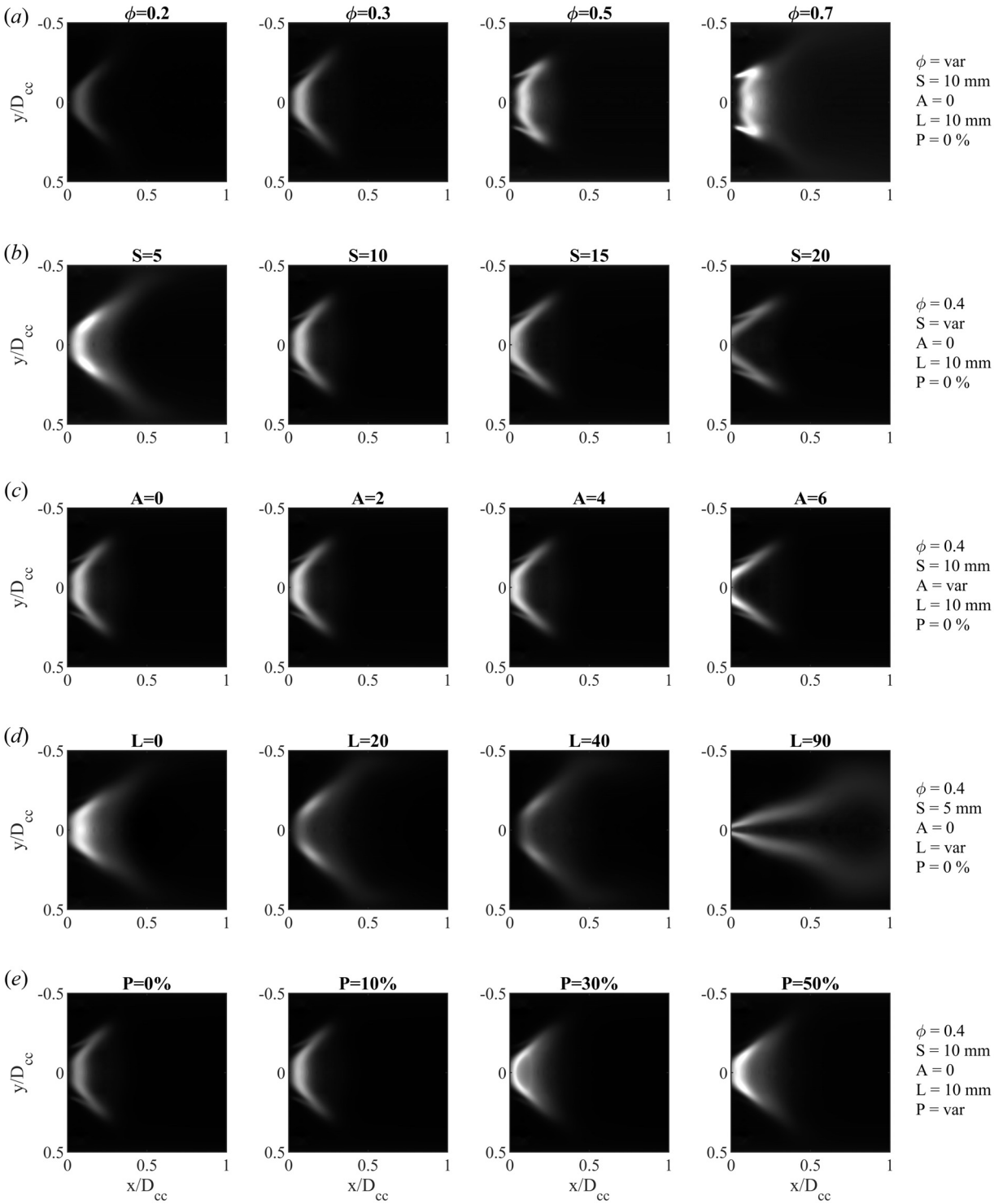

Fig. 11 Parameter study of deconvoluted OH* intensities with variations of (*a*) the equivalence ratio at constant thermal power, (*b*) the swirler blockage *S*, (*c*) the axial air restriction *A*, (*d*) the center body position *L*, and (*e*) the share of central fuel injection *P*

hydrodynamic effect is slightly dominant regarding the flame position in this case because the flame root on the centerline is pushed downstream with the increasing equivalence ratio. Hence, the richer flames are not more prone to flashback than the lean ones. A second effect can be observed in the outer shear layers. The flame transitions from V to M shape [52] because the significantly lower air mass flow (Fig. 3) generates much lower strain rates in the SL. Furthermore, the flame becomes more compact from $\phi = 0.5$ to 0.7 due to a combination of the lower flow velocities and the increased flame speed.

Figure 11(*b*) shows the influence of the swirl intensity on the flame shape at a constant equivalence ratio. For moderate swirl intensities, the flame stabilizes inside the combustion chamber. Between $S = 5$ mm and $S = 10$ mm it, transitions from V to M shape without changing the flame root position. Further increasing the swirl enhances the VD in the mixing tube and shifts the RZ upstream (Fig. 7), which enables the flame to propagate upstream as well. At $S = 20$ mm, the flame root lies completely inside the mixing tube. The flame is still stable but has a low robustness against flashback as shown in Fig. 10.

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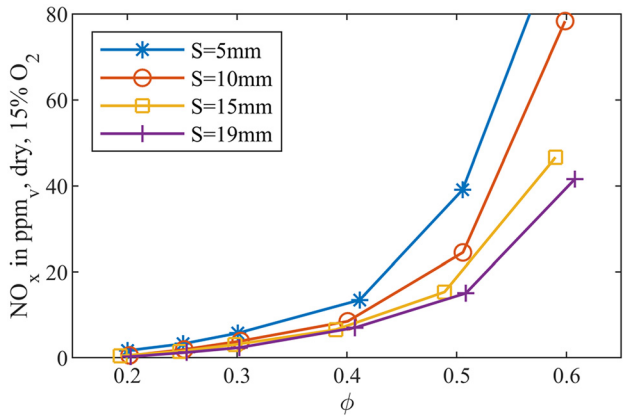

Fig. 12 NO_x emissions for different swirl intensities and $A = 0$, $L = 10$ mm, and $P = 0\%$ with standard normalization (Eq. (1))

The effect of restricting the axial air injection (A) on the flame shape is similar to increasing the swirl intensity because the VD is enhanced as well (Fig. 11(*c*)). However, since the swirl is nearly constant for all images, the OH* intensity at $A = 6$ appears more narrow than for the high swirl flame.

The variation of the fuel lance position shows two different behaviors. The low-swirl V shape flame, depicted in Fig. 11(*d*), has the highest stability against flashback for a variation of L, as stated before in Sec. 5. Up to $L = 50$ mm, the flame shape and position stays nearly constant, but the intensity distribution is shifted from the center to the SL. This effect can be contributed to the change in the fuel concentration profile shown in Fig. 8. At $L > 50$ mm, the center of the flow begins to lean out because the fuel is no longer sufficiently mixed into the wake of the lance (at $P = 0$). As the fuel lance approaches the end of the mixing tube, the characteristic shape of a swirl-stabilized flame disappears. Instead, the flame burns in the shear layers of the wake flow with a very narrow V-shape. The flow does not appear to undergo vortex breakdown anymore due to the heat-driven expansion of the fluid. In the case of higher swirl intensity ($S \geq 10$ mm), the M-shaped flame is affected in a similar manner as for the restriction of the axial air injection in Fig. 11(*c*) because the wake of the fuel lance also amplifies the VD. As a consequence, the flame exhibits flashback if the lance position exceeds a certain value depending on the swirl intensity (Fig. 10).

Increasing the central fuel injection (P) shifts the H_2 distribution toward the centerline as shown in Fig. 8. Hence, the local equivalence ratio rises in the middle and declines near the walls of the mixing tube. The flame transitions from M- to V-shape and the OH* intensity at the flame root increases significantly.

All of the varied parameters in this section have a clear impact on the flame topology. Hence, they qualify to be used as design parameters in the planned optimization process. Moreover, constraints for the parameter space can be derived from the recorded flame shapes and the stability map in Fig. 10. A second important aspect is the influence of the parameters on the emissions, which will be discussed in Sec. 7.

7 Emissions

The strongest impact on the NO_x emissions has naturally the variation of the equivalence ratio as depicted in Fig. 12. The emissions rise for all curves due to the increasing flame temperature. For some design points, single digit NO_x can be reached up to $\phi = 0.4$.

Figure 12 presents also the influence of the swirl parameter S, which reveals a monotonic behavior as increased swirl intensity reduces the NO_x emissions significantly. On the one hand, the flame is very compact at low swirl, while it is stretched with increasing swirl intensity (Fig. 11(*b*)), which improves heat release distribution and could reduce NO_x. On the other hand, the higher swirl intensity increases the turbulence in the mixing tube and leads to a more homogeneous air/fuel mixture in both the spatial and temporal domain. This effect was measured for a similar burner by Reichel et al. [45] and reduces local and temporary hot spots in the flame. Since the NO_x emissions rise exponentially with the flame temperature, such hot spot related fluctuations cause increased NO_x production. The combination of these effects halves the NO_x values from $S = 5$ mm to $S = 19$ mm, which is a remarkable reduction. In the extreme case at $S = 19$ mm, Fig. 11(*b*) shows very high stretching of the flame. The high strain rate could possibly lead to local quenching, which would also reduce the NO_x emissions. However, the exhaust H_2 concentrations and temperatures show no signs of incomplete combustion in this case.

The effect of the fuel distribution is further investigated by shifting H_2 to the fuel lance, i.e., increasing P, and, hence, artificially impairing the fuel/air homogeneity. Figure 13 shows the influence of a single parameter on the emissions while the other parameters are held constant. Shifting the fuel to the central injection (P) increases the local equivalence ratio in the center of the flow. As expected, the emissions increase accordingly. Interestingly, for $P \geq 20\%$ the curve starts to saturate. Even though, the fuel distribution (Fig. 8) and flame shape (Fig. 13) are still changing, the maximum flame temperature seems not to rise much further. Please note that the fuel lance is placed at $L = 10$ mm for this curve, which means that the point of central injection is 20 mm downstream of the premixing nozzles and 80 mm upstream of the combustion chamber. Hence, the flame can still be regarded as technically premixed rather than diffusive.

The variation of the lance position (L) has no effect on the emissions as long as $P = 0\%$. Restricting the axial air injection (A) has also a low influence on the emissions up to the point close to flashback. At $A = 6$ the NO_x increase slightly. This is caused by a strong narrowing and, hence, concentration of the flame (Fig. 11(*c*)). Even though both parameters show a low influence on the emissions individually, they might combine with other parameters as they still alter the flame and should be included in the later optimization nonetheless.

Please note that the absolute values of Figs. 12 and 13 cannot be compared directly with each other because they were measured in two campaigns with slightly different setups (e.g., passive/no cooling). Nevertheless, both plots are consistent individually, which justifies the statements made.

Another aspect of the optimization must be the pressure loss of the burner because by integrating the combustor into the mGT, its pressure loss is limited to the capabilities of compressor and turbine. Therefore, the pressure in the plenum was monitored in difference to the ambient pressure for all emission measurement points. Figure 14 depicts the relationship between both quantities for different burner

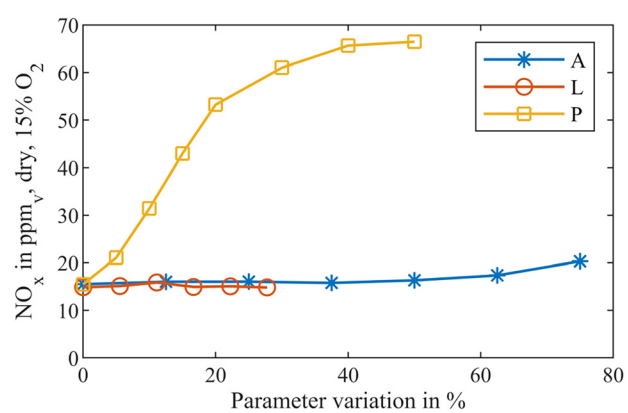

Fig. 13 NO_x emissions dependence of the axial air restriction (A), the lance position (L) and the central fuel injection (P). The parameter variation in % relates to the full parameter spaces: $A = [0, 8]$, $L = [0, 90]$ mm and $P = [0, 100]\%$. The series were measured at $\phi = 0.48$ and plotted with standard normalization.

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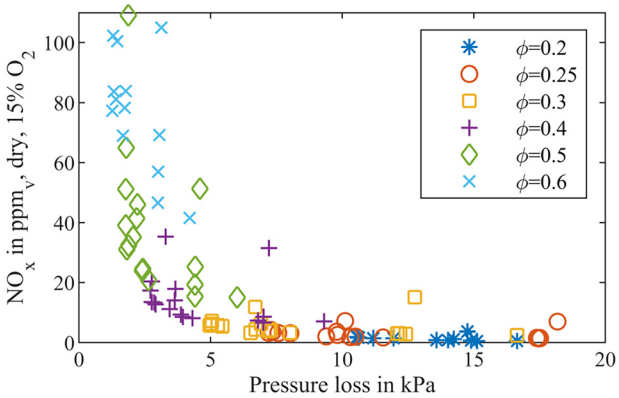

Fig. 14 NO_x emissions over pressure loss for various burner configurations and equivalence ratios

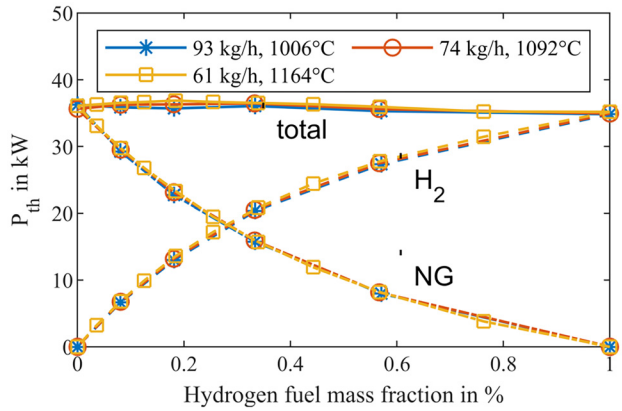

Fig. 15 Thermal power of the fuel mixtures with the H_2 (dashed) and natural gas (dash-dotted) share at three different air mass flows and exhaust temperatures

Table 2 Composition of the natural gas

Gas		Mole fraction
Methane	CH_4	89.42
Nitrogen	N_2	1.55
Carbon dioxide	CO_2	1.50
Ethane	C_2H_6	5.83
Propane	C_3H_8	1.28
Butane	C_4H_{10}	0.33
Rest	C_5+	0.09

configurations and equivalence ratios. It reveals a pareto front, which is based on the main influence parameters ϕ and S. As shown before, the emissions rise with increasing ϕ and decreasing swirl intensity. At the same time, the pressure loss decreases with rising ϕ because the equivalence ratio is controlled via the air mass flow. The lower mass flow through the burner results in less pressure loss. The swirl intensity is controlled by blocking the swirl generator slots. Hence, the effective passage surface is reduced and the higher velocity increases the pressure loss. In the optimization, ϕ won't be a free parameter because it is determined by the splitting ratio of primary and secondary air. For S on the other hand, the impact on the pressure loss must be regarded either in the form of a constraint or a multi-objective optimization.

8 Fuel Mixtures

For the initial tests at machine level, it might be necessary to start with natural gas and increase the H_2 content gradually, in case full H_2 burning causes flashbacks or other unexpected events. Therefore, the burner is tested in the test rig with natural gas as well. The H_2 and natural gas (Table 2) are mixed in the supply line to the burner with respect to a constant thermal power level as it is shown in Fig. 15. Furthermore, the air mass flow is fixed for each measurement series, which also means that the exhaust temperature stays approximately constant. With this procedure, the burner allows switching the fuel from 100% natural gas to 100% H_2. However, since the burner was designed for the properties of H_2, the stability range of the natural gas flame is reduced significantly from $\dot{m}_{\mathrm{air,max}} > 200$ kg/h to $\approx$ 100 kg/h due to flame blow off.

Figure 16 presents the OH* intensity of flames featuring different H_2 contents in natural gas. The pure natural gas flame reveals a very long V-shape almost touching the combustion chamber walls. With rising H_2 content, the flame becomes more compact due to the increased reactivity. Finally, the flame turns into M-shape as seen in Sec. 6, Fig. 11.

The evolution of the NO_x emissions during this transition is of particular interest. As discussed in the introduction, the addition of H_2 to gas turbines is often seen to increase NO_x emission. Please note that the equivalence ratio of each series is not constant because the stoichiometry of the fuel changes with the H_2 mass fraction. Instead, the air mass flow and the exhaust temperature are kept constant. The total mass flow of the exhaust reduces only by $2-2.5\%$ and the volume flow increases by $1-1.5\%$ for H_2 compared to natural gas. Hence, the properties relevant for the turbine are all nearly constant for this comparison.

Figure 17 shows a slight rise of the NO_x emissions with the H_2 content indeed. However, this rise cannot be seen in the raw data, but is imposed by the standard normalization to dry exhaust gas with 15% oxygen content. This normalization is tailored to the regular combustion of natural gas or methane and is intended to compensate variations of the global equivalence ratio. Thereby it assumes an otherwise relatively constant composition of the exhaust gas. This is not the case for a fuel switch. The substitution of natural gas by H_2

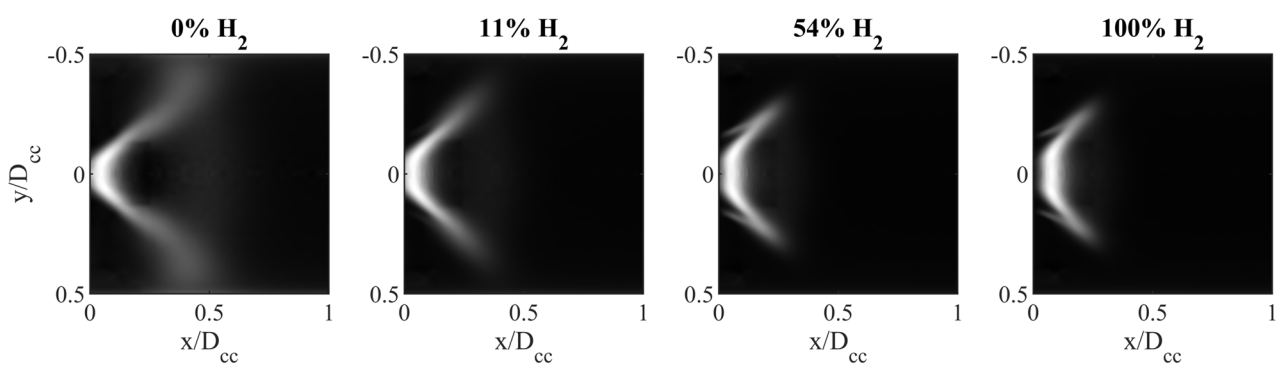

Fig. 16 Deconvoluted OH* intensity of premixed flames ($\dot{m}_{\mathrm{air}}$=93 kg/h) blending from natural gas to H_2 in mass fraction. Please note that the colorbars were individually adjusted for better visibility.

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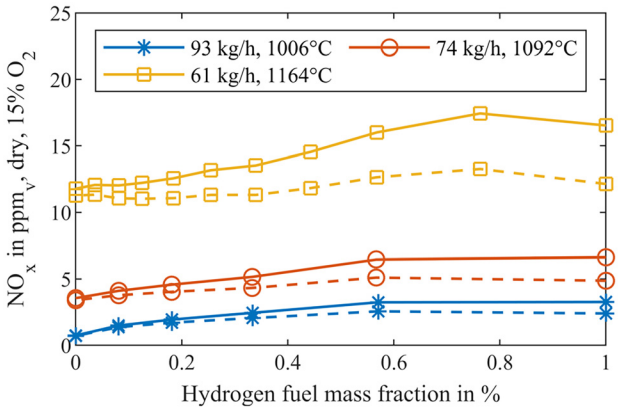

Fig. 17 NO_x of H_2/natural gas mixtures at three different air mass flows and exhaust temperatures with standard (solid, Eq. (1)) and alternative normalization (dashed, Eq. (2))

leads to a higher water and oxygen content because the exhaust does not contain CO_2 and H_2 needs less oxygen for similar heat release. In the standard normalization, these elevated concentrations cause a higher correction coefficient

$$c_s = \frac{1}{1 - X_{H_2O}} \frac{0.21 - X_{O_2,\mathrm{ref}}}{0.21 - X_{O_2}} \tag{1}$$

with $X_{O_2,\mathrm{ref}} = 0.15$. Douglas et al. [53] stated in a white paper that this artificial increase equals about 37% for pure H_2 compared with methane, even though the same mass NO_x is produced. This difference can also be seen in Fig. 17 between the solid and the dashed lines. The latter use a normalization method proposed by Garan et al. [54], which takes the whole exhaust gas composition into account

$$c_a = \frac{1}{1 - X_{H_2O}} \frac{0.21 - X_{O_2,\mathrm{ref}}}{0.21 - X_{O_2}} \frac{\alpha + 1 - 0.21}{\alpha_{\mathrm{ref}} + 1 - 0.21} \frac{\beta'}{\beta'_{\mathrm{ref}}} \tag{2}$$

with the fuel-specific parameters

$$\alpha = \frac{n_C^f + n_S^f + n_{N_2}^f}{\left(n_H^f - n_{CL}^f\right)/4 - n_{O_2}^f - n_{N_2}^f} \tag{3}$$

and

$$\beta' = \frac{\beta}{\mathrm{LHV}} = \frac{\left(n_H^f - n_{CL}^f\right)/4 - n_{O_2}^f - n_{N_2}^f}{\mathrm{LHV}}. \tag{4}$$

Here, each n_i^f represents the amount of element or species i within each fuel molecule. We chose methane as reference gas such that $\alpha_{\mathrm{ref}} = \beta_{\mathrm{ref}} = 1$ (see Table 3). This normalization method almost completely compensates the incline of the NO_x emissions when increasing the H_2 content (Fig. 17). The overall differences now fall in the order of the measurement uncertainty of ± 1 ppm. This result indicates that there is no significant influence of the H_2 enrichment

Table 3 Fuel properties

Property	H_2	Natural gas	CH_4 ref	Unit
LHV mass	119.97	46.5	50.01	MJ/kg
LHV volume	10.78	37.51	35.8	MJ/m^3
Molar mass	2.016	—	16.04	g/mol
Wobbe index	40.90	47.52	48.17	MJ/m^3
α	0	—	1	
β	0.5	—	1	

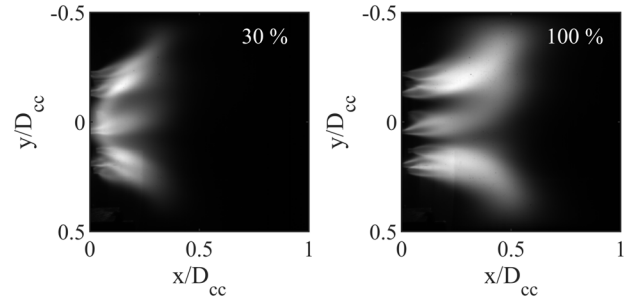

Fig. 18 Line-of-sight OH* intensity of a diffusion H_2 flame with 30% (l.) and 100% (r.) front plate injection

on the NO_x emissions, which should be further investigated in the future.

9 Flashback Recovery

The parameter study in Sec. 5 showed a large stable operation window for the burner, which should allow safe operation in the mGT. However, the flame properties, e.g., the flame speed, change during the ramp-up of the static pressure (to 3.6 bara). The risk of flashbacks can be particularly high during certain transient conditions such as startup or load changes.

The burner features eight additional fuel nozzles at the burner front plate that can be used as an inherently stable pilot flame. The individual diffusion flames from each nozzle can be seen clearly in Fig. 18 (r.), which shows the OH* intensity of the flame at 100% pilot. As soon as flashback is detected, the H_2 can be shifted to the pilot. The burner can be brought back to stable conditions before the fuel is shifted to the premixed mode again. This procedure has been tested successfully in the test rig by triggering flashback via the fuel lance position.

The front plate pilot should only be used shortly for flashback recovery as the NO_x emissions are much higher than with premixing. Figure 19 presents a comparison between the full (technically) premixed mode, the fuel lance and the front plate pilot. As discussed before in Fig. 13, the emission rise for shifting H_2 to the fuel lance saturates at $P = 40\%$. The emissions of the front plate pilot, on the other hand, increase continuously. However, the NO_x curve of this flame increases much slower than the one with the fuel lance until 70% pilot share. This is surprising since the injection via the front plate is pure H_2 that ignites immediately, while the injection via the fuel lance is done approximately 80 mm upstream of the flame. Hence, the fuel homogeneity should be better using the fuel lance, but the emissions are still higher for low and medium P.

The OH* recordings of the front plate pilot at 10–40% show a superposition of the premixed flame and a short pilot flame (Fig. 18

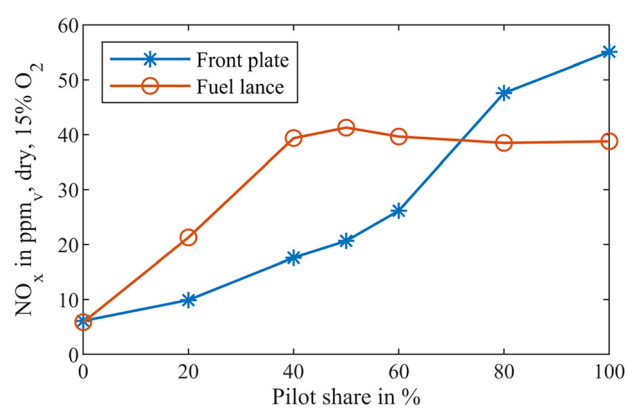

Fig. 19 Evolution of the NO_x emissions (with standard normalization) when switching from premixed to front plate pilot or fuel lance ($L = 10$ mm) at ϕ=0.4

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l.). Afterwards, the premixed flame is not visible anymore like in the right image of Fig. 18. An explanation for the slow increase of the NO_x emissions during the transition from premixed to the front plate pilot might be found in a different H_2 combustion technology. Funke et al. [55] present an overview on micromix combustors, which utilize miniaturized diffusion flames to mitigate the flashback risk of H_2. It is reported that these flames, even though they do not have a premixing section, feature low NO_x emission levels due to the reduced residence time in the flame region [55]. A similar effect might be present here as long as the H_2 mass flow through the front plate nozzles is low. From 60% onward, the pilot flames get significantly longer and the emissions increase more rapidly. At full piloting, the diffusion flames emit more NO_x than the partially premixed flame from the fuel lance.

10 Conclusion

The paper presents the results of the first phase of the H2mGT project, which aims to develop and validate a hydrogen burner for a micro gas turbine. It is shown that the burner can be stably operated for multiple parameter settings with 100% hydrogen fueling. The influence of four geometric parameters on the flame shape is presented. At certain design points, flashback is triggered by an amplification of the velocity deficit on the centerline of the mixing tube. This amplification is caused by the position of the fuel lance and its wake, the restriction of axial air injection, and increased swirl intensity. At the same time, high swirl intensity lowers the emissions and indeed single digit NO_x can be reached (up to $\phi = 0.4$ at atmospheric pressure). Next to the equivalence ratio, the emissions are mainly influenced by the fuel injection strategy and the swirl intensity in the mixing tube. Switching the fuel from natural gas to hydrogen at constant geometry alters the flame shape but does not lead to a significant increase in emissions. However, the standard volumetric normalization to 15% O_2 artificially overemphasizes the NO_x by almost 40%. This normalization is often used in governmental regulations, e.g., the German TA Luft, and penalizes the usage of hydrogen. Hence, such regulations should be adjusted in the future.

In the next phase of the project, the burner is equipped with a counterflow cooled steel combustion chamber and secondary air injection. Using this setup, an optimization toward low emissions and high flame stability is performed. The results from the first phase are used to define the parameter space of the optimization. The variation of parameters showed a certain correlation between the emissions and the burner pressure loss. As the possible pressure loss of the burner in the mGT is limited by the compressor and turbine capabilities, it needs to be constrained in the optimization.

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Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

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